

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SAVEETHA COMPILER ASSESSMENT (LEVEL 1 & VIVA VOCE)

Course Code:	CSA09	Course Title:	Programming in Java
CO Assessed:	CO1 – Manipulation (BL4)	Assessment:	AT1 – Saveetha Compiler (Level 1 & Viva)
Weightage:	20% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	CO1: <i>Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3, BL4)</i>		
Student Name:	L. Sanjay Kumar	Reg. No.:	192425226
Section / Batch:	AIML/2024	Date:	27-07-2026

Instructions to Students

Answer ALL 5 questions. Total: 100 Marks.

Questions 1–5: Write complete, syntactically correct Java programs on the Saveetha Compiler with proper indentation. Each question carries 20 Marks.

For program questions: test with at least two sample inputs; ensure output is displayed clearly. Each question carries 5 Marks.

No calculators or electronic devices permitted during the viva session.

Question Type	Description	Questions	Marks
Level 1 – Program Writing	Control Flow, classes and Object programs, Collection Framework on compiler	Q1 – Q4	50
GRAND TOTAL:			100 Marks

SECTION A – LEVEL 1: PROGRAM WRITING QUESTIONS (Q1–Q5)

Q1. (Level 1 – Program Writing) [20 Marks]

Print Numbers from M to N by Skipping K Numbers in the Middle

START

Input A

Input B

Input C

$\text{mid} \leftarrow (A + B) / 2$

$\text{startSkip} \leftarrow \text{mid} - (C / 2)$

FOR $i \leftarrow A$ TO B

IF $i \geq \text{startSkip}$ AND $i < \text{startSkip} + C$

CONTINUE

END IF

Print i

END FOR

STOP

Answer / Program with output:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, a 'Secure Exit' button, the lab name 'SIMATS Online Programming Lab', the subject 'JAVA PROGRAMMING', and a user profile for 'LENKA SANJAY KUMAR' with ID '192425226RA'.

On the left, a sidebar lists four questions: Q1 (Numbers), Q2 (Loops), Q3 (Strings), and Q4 (Array). All are marked as '4 / 4 submitted'.

The main area contains the problem statement: 'Write a program to print the numbers from M to N by skipping K numbers in between?'. It provides sample input (M=50, N=100, K=7) and sample output (50, 58, 66, 74, ...). Below this, five test cases are listed:

1. M = 15, N = 05, K = 02
2. M = 25, N = 50, K = 04
3. M = 15, N = 100, K = -02
4. M = 0, N = 0, K = 2
5. M = 200, N = 200, K = 50

The 'Your Code' section shows a Java program:

```
1 import java.util.Scanner;
2 public class num
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int a,b,c;
8         a=sc.nextInt();
9         b=sc.nextInt();
10        c=sc.nextInt();
11        for(int i=a;i<b;i++)
12        {
13            i=i+c;
14            System.out.println(i);
15        }
16    }
17 }
```

The 'Output' section shows the result of the program for the sample input:

```
3
5
7
9
```

Q2. (Level 1 – Program Writing) [20 Marks]

Factorial of a Number

START

Input n

fact $\leftarrow$ 1

FOR i $\leftarrow$ 1 TO n

 fact $\leftarrow$ fact $\times$ i

END FOR

Print fact

STOP

Answer / Program with output:

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On the left, a sidebar lists four questions: Q1 (Numbers), Q2 (Loops), Q3 (Strings), and Q4 (Array), all marked as completed. Below this, it indicates '4 / 4 submitted'.

The main workspace is titled 'Find the factorial of n?'. It provides a 'Sample Input: N = 4' and a 'Sample Output: 4 Factorial = 24'. Below the sample output, a list of test cases is shown: 1. N = 0, 2. N = -5, 3. N = 1, 4. N = Q, and 5. N = 3A.

The 'Your Code' section contains the following Java code:

```
1 import java.util.Scanner;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int n,fact=1;
8         n=sc.nextInt();
9         for(int i=1;i<=n;i++)
10        {
11            fact=fact*i;
12        }
13        System.out.println("Factorial = "+fact);
14    }
15 }
16
17
```

The 'Output' section shows the result: 'Factorial = 5040'.

Q3. (Level 1 – Program Writing) [20 Marks]

Find a Character in a String

START

Input string

Input character

index $\leftarrow$ -1

FOR each character in string

IF current character = input character

index $\leftarrow$ current position

EXIT loop

END IF

END FOR

IF index $\neq$ -1

Print "Character found at index", index

ELSE

Print "Character not found"

END IF

STOP

Answer / Program with output:

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On the left, a sidebar lists four questions: Q1 Numbers, Q2 Loops, Q3 Strings (selected), and Q4 Array. Below this, it indicates '4 / 4 submitted'.

The main area contains the problem statement: 'Write a program that finds whether a given character is present in a string or not. In case it is present it prints the index at which it is present. Do not use built-in find functions to search the character.' It provides sample input and output, and a note to check for non-available characters in hidden test cases.

The 'Your Code' section shows the following Java code:

```
1 import java.util.Scanner;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         String s=sc.nextLine();
8         char a=sc.next().charAt(0);
9         int ind=-1;
10        for(int i=0;i<s.length();i++)
11        {
12            if(s.charAt[i]==a)
13            {
14                ind=i;
15                break;
16            }
17        }
18        if(ind!=-1)
19            System.out.println("character found at index :");
20        else
21            System.out.println("not found");
22    }
23 }
```

The 'Output' section shows a 'Compilation Error' message: 'C:\Windows\TEMP\sandbox_192425226RA_6a671dea4fd9b4.65661333\main.java:12: error: cannot find symbol if(s.charAt[i]==a)'.

Q4. (Level 1 – Program Writing) [20 Marks]

Pseudocode for Mean, Median, and Mode

START

Input n

Create array of size n

sum $\leftarrow$ 0

FOR i $\leftarrow$ 0 TO n-1

 Input array[i]

 sum $\leftarrow$ sum + array[i]

END FOR

mean $\leftarrow$ sum / n

Sort the array

IF n is even

 median $\leftarrow$ (array[n/2 - 1] + array[n/2]) / 2

ELSE

 median $\leftarrow$ array[n/2]

END IF

mode $\leftarrow$ array[0]

maxCount $\leftarrow$ 0

FOR i $\leftarrow$ 0 TO n-1

 count $\leftarrow$ 1

 FOR j $\leftarrow$ i+1 TO n-1

 IF array[i] = array[j]

 count $\leftarrow$ count + 1

 END IF

 END FOR

 IF count > maxCount

 maxCount $\leftarrow$ count

 mode $\leftarrow$ array[i]

 END IF

END FOR

Print "Mean = ", mean

Print "Median = ", median

Print "Mode = ", mode

STOP

Answer / Program with output:

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LEVEL 1 — EASY

- Q1 Numbers ✓
- Q2 Loops ✓
- Q3 Strings ✓
- Q4 Array ✓

4 / 4 submitted

Find the Mean, Median, Mode of the array of numbers?
Sample Input::
Array of elements = {16, 18, 27, 16, 23, 21, 19}
Sample Output:
Mean = 20
Median = 19
Mode = 16

1. Array of elements = {26, 28, 37, 26, 33, 31, 29}
2. Array of elements = {1.6, 1.8, 2.7, 1.6, 2.3, 2.1, .19}
3. Array of elements = {0, 160, 180, 270, 160, 230, 210, 190, 0}
4. Array of elements = {200, 180, 180, 270, 160, 270, 270, 190, 200}
5. Array of elements = {100, 100, 100, 100, 100, 100, 100, 100, 100}

Your Code

```
1 import java.util.Scanner;  
2 public class main  
3 {  
4     public static void main()  
5     {  
6         Scanner sc=new Scanner(System.in);  
7         System.out.print("enter size");  
8         int n=sc.nextInt();  
9         int sum=0;  
10        int[] a= new int[n];  
11        for(int i=0;i<n;i++)  
12        {  
13            a[i]=sc.nextInt();  
14            sum+=a[i];  
15        }  
16        double mean=sum/n;  
17        System.out.println("mean = "+mean);  
18        double median;
```

Output

```
Error: Main method not found in class  
main, please define the main method as:  
    public static void main(String[] args)  
or a JavaFX application class must extend  
javafx.application.Application
```

RUBRICS FOR EVALUATION

Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (3–4 Marks)	Level 2 – Developing (2 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Analysis & Understanding				
Logic / Algorithm Design				
Python Syntax & String Operations				
Output Correctness & Testing				
Communication & Explanation (Viva)				

Code of Conduct

I L. Sanjay Kumar/192425226 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: L.S.Kumar